

C++ MINI PROJECT PLAN

HOSTEL MANAGEMENT SYSTEM

PERSON2:251136:BACKEND

PLAN:

Backend Development Plan

1. Technology Selection

- C++
- MySQL
- MySQL Connector/C
- cpp-httplib
- nlohmann/json
- C++ OOP concepts

2. Database Connection

- Connect the C++ backend with the MySQL database.
- Test whether the connection is successful.
- Create basic functions for executing SQL queries.

3. Student Management

- Create a Student class.
- Add student records.
- View student records.
- Update student details.
- Delete student records.

4. Room Management

- Create a Room class.
- Add room details.
- View available rooms.
- Track room capacity and vacancies.
- Update room information.

5. Room Allocation

- Create an Allocation class.
- Assign students to rooms.
- View allocation details.
- Change or remove room allocations.
- Update room vacancy after allocation.

6. Complaint Management

- Create a Complaint class.
- Allow complaints to be stored in the database.
- View complaints.
- Update complaint status.
- Delete complaints when required.

7. Login System

- Implement a basic login system.
- Verify username and password from the database.
- Provide different access for students and administrators.

8. Backend API

- Use `cpp-http-lib` to create HTTP endpoints.
- Connect the frontend requests with backend functions.
- Return data in JSON format using `nlohmann/json`.

- Create APIs for students, rooms, allocations, complaints, and login.

9. Backend Testing

- Test database connection.
- Test CRUD operations for each module.
- Test room allocation and vacancy updates.
- Test complaint operations.
- Test login.
- Test all API endpoints.

10. Backend Documentation

- Document the backend structure.
- Document database operations.
- Document API endpoints.
- Record testing results.
- Keep the code simple and understandable so that every module can be explained during the project evaluation.